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Performance of an algorithms predicting mortality in medical vs surgical patient with diabetes

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Introduction: Patients with diabetes have an increased risk of hospital complications including complications related to surgical treatment. Computer-based early warning systems (EWS) implemented in the hospital wards can identify patients at risk of deterioration, and trigger lifesaving interventions. In general, these systems often perform differently in subpopulations of patients. There is also concern with risk factors associated with chronic conditions that are usually not part of the variables used by these algorithms. We present preliminary data of the performance of an EWS in the general hospital wards in patients with diabetes assigned to medical vs surgical treatment.

Methods: Adult (≥ 18 y-o) patients with diagnosis of diabetes, hospitalized in general wards were grouped based on medical vs. surgical treatment. We collected EWS scores generated every 15 min during the entire hospitalization. The scores estimated the probability of total mortality on scale 0 to 100. We compared the distributions of the Highest Score (at any time during the hospitalization) and calculated diagnostic accuracy using a cutoff of ≥ 60 .

Results: There were 15,727 patients with diabetes, 10,234 in the medical group (age mean(SD) 68.8(13.9) years; female 44%; length of stay 5.0(5.7) days) and 5,493 in the surgical group (age mean(SD) 67(12.7) years; female 40%; length of stay 6.3(8.9) days). Hospital mortality in the medical vs. surgical group was 88 (0.86%) vs. 6 (0.1%). Using a score of ≥ 60 to predict mortality, the model had an accuracy in the medical group of 85% (95%CI: 84.44 to 85.83) and Sens 89.78%, Spec 85.11%, PPV 4.97% and NPV 99.9%. In the surgical group the accuracy was 81.83% (95%CI: 80.79 to 82.84) and Sens 66.67%, Spec 81.85%, PPV 0.40% and NPV 99.96%.

Conclusion: The performance of the EWS model predicting mortality was different among patients with diabetes admitted for medical vs. surgical treatment. The overall accuracy of the model is statistically better in the medical group with Sens and Spec above 85%; while in the surgical group the Sens is quite low at 66%. The NPV in both groups is clinically relevant at more than 99%, but the PPV is extremely low. The difference among these subpopulations should be validated with other EWS and should be considered when using the predictions of these models in medical decision making. Our results should be interpreted cautiously, given they are from a single institution with a fairly low prevalence of adverse events including

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